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Report of

First-order methods for constrained convex programs

Author
Dong Hu

Supervisor
Prof. Yangyang Xu

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1 Introduction

This report will be mainly discuss about the application of the alternating direction method of multipliers(ADMM). It takes the form of a decomposition-coordination procedure, in which the solutions to small local subproblems are coordinated to find a solution to a large global problem. ADMM can be viewed as an attempt to blend the benefits of dual decomposition and augmented Lagrangian methods for constrained optimization.

Under the help with professor Yangyang, this report will show two basic applications of the ADMM method: Basis Pursuit(BP) and Least Absolute Deviations(LAD)

2 Application and Results

2.1 Alternating Direction Method of Multipliers

The algorithm solves problems in the form:

$$\begin{array}{ll} \text{minimize} & f(x) + g(z) \\ \text{subject to} & Ax + Bz = c \end{array}$$

with $x \in R^n$ and $z \in R^m$, where $A \in R^{p \times n}$, $B \in R^{p \times m}$, and $c \in R^p$. Also, we define the form the augmented lagrangian

$$L_\rho(x, z, y) = f(x) + g(z) + y^T(Ax + Bz - c) + \frac{\rho}{2}||Ax + Bz - c||_2^2$$

From the above fomula, it can be seen that we can split the primal variable and the objective function, but no longer like the dual ascent method, which the splitted x_i should be regarded as part of x, and needs to be merged together afterwards. What ADMM do is to treat the disassembled variables as different variables x and z at the start, and so does the constraints. The advantage is that there is no need to combine x and z together, and that guarantees The decomposability of the previous optimization process.

Therefore, ADMM consists of the iterations:

$$\begin{aligned} x^{k+1} &\leftarrow \arg\min_x L_p(x, z^k, y^k) \\ z^{k+1} &\leftarrow \arg\min_z L_p(x^{k+1}, z, y^k) \\ y^{k+1} &\leftarrow y^k + \rho(Ax^{k+1} + Bz^{k+1} - c) \end{aligned}$$

For convenience, we can combine the linear and quadratic terms in the augmented Lagrangian and scaling the dual variable. If we define the residue $r = Ax + Bz - c$ and scaled dual variable $u = \frac{1}{\rho}y$, we have

$$\begin{aligned} y^T r + \frac{\rho}{2}||r||_2^2 &= \frac{\rho}{2}||r + \frac{1}{\rho}y||_2^2 - \frac{1}{2\rho}||y||_2^2 \\ &= \frac{\rho}{2}||r + u||_2^2 - \frac{\rho}{2}||u||_2^2 \end{aligned}$$

Defining the residual at iteration k as $r^k = Ax^k + Bz^k - c$, we can see that the running sum of the residuals is

$$u^k = u^0 + \sum_{j=1}^{k} r^j$$

then we can express the simplified ADMM form as

$$\begin{aligned}
x^{k+1} &\leftarrow \arg\min(f(x) + \frac{\rho}{2}||Ax + Bz^k - c + u^k||_2^2) \\
z^{k+1} &\leftarrow \arg\min(g(z) + \frac{\rho}{2}||Ax^{k+1} + Bz - c + u^k||_2^2) \\
u^{k+1} &\leftarrow u^k + Ax^{k+1} + Bz^{k+1} - c
\end{aligned}$$

Although the original form is ready to solve the problem, but usually the simplified form will be easier to solve the applications in the future. And surly the form is used for LAD and BP in the following two section.

2.2 Basis Pursuit

Description:

$$\begin{array}{ll} \text{minimize} & ||x||_1 \\ \text{subject to} & Ax = b \end{array}$$

with $x \in R^n$, $A \in R^{m \times n}$, $b \in R^m$, with $m < n$.

Equivalently, we can write the question in another form:

$$\begin{array}{ll} \text{minimize} & ||x||_1 \\ \text{subject to} & Az = b \\ & x = z \end{array}$$

Then if we write the constraints in vector form, we have:

$$\begin{bmatrix} I \\ 0 \end{bmatrix} x + \begin{bmatrix} -I \\ A \end{bmatrix} z = \begin{bmatrix} 0 \\ b \end{bmatrix}$$

So the augmented lagrangian is:

$$L_\rho(x, z, u_1, u_2) = ||x||_1 + < u_1, Az - b > + < u_2, x - z > + \frac{\rho}{2}||Az - b||_2^2 + \frac{\rho}{2}||x - z||_2^2$$

According to Scaled ADMM algorithm we derived in (2.1), x^{k+1} for basis pursuit will be:

$$x^{k+1} \leftarrow \arg\min_{x \in R^n}(||x||_1 + < u_2^k, x - z > + \frac{\rho}{2}||x - z^k||_2^2) \tag{2.1}$$

$$= \arg\min_{x \in R^n}(||x||_1 + \frac{\rho}{2}||x - z^k + \frac{u_2^k}{\rho}||_2^2 - \frac{1}{2\rho}||u_2^k||_2^2) \tag{2.2}$$

Since the last term $-\frac{1}{2\rho}||u_2^k||_2^2$ is irrelevant to x, we have

$$x_i^{k+1} \leftarrow \arg\min_{x_i}(||x_i||_1 + \frac{\rho}{2}||x_i - z_i^k + \frac{u_2^k}{\rho}||_2^2) \tag{2.3}$$

there are three cases

$$x_i^{k+1} = \arg\min_{x_i^k}(|x_i^k| + \frac{\rho}{2}(x_i^k - z_i^k + \frac{u_2^k}{\rho})^2) = \begin{cases} 0 & \text{if } |z_i^k + \frac{u_2^k}{\rho}| \leq \frac{1}{\rho} \\ z_i^k + \frac{u_2^k}{\rho} - \frac{1}{\rho} & \text{if } \quad z_i^k + \frac{u_2^k}{\rho} > \frac{1}{\rho} \\ z_i^k + \frac{u_2^k}{\rho} + \frac{1}{\rho} & \text{if } \quad z_i^k + \frac{u_2^k}{\rho} < -\frac{1}{\rho} \end{cases}$$

Also,z^{k+1} for basis pursuit will be:

$$z^{k+1} \leftarrow \arg\min_{z \in R^n}(< u_1^k, Az - b > + < u_2^k, x^{k+1} - z > + \frac{\rho}{2}||Az - b||_2^2) + \frac{\rho}{2}||x^{k+1} - z||_2^2) \tag{2.4}$$

To get the argmin, we have the gradient equation to solve:

$$0 = A^T u_1^k - u_2^k + \rho A^T(Az - b) + \rho(z - x^{k+1})$$

After that, we solve the linear equations of u_1^{k+1} and u_2^{k+1}:

$$\begin{cases} u_1^{k+1} & \leftarrow u_1^k + \rho(Az^{k+1} - b) \\ u_2^{k+1} & \leftarrow u_2^k + \rho(x^{k+1} - z^{k+1}) \end{cases}$$

and from that, along with iteration (2.2) and (2.4) completes the ADMM for BP.

2.3 Least Absolute Deviations

Description:

$$\begin{array}{ll} \text{minimize} & ||z||_1 \\ \text{subject to} & Ax - z = b \end{array}$$

The augmented lagrangian is:

$$L_\rho(x, z, u) = ||z||_1 + < u, Ax - z - b > + \frac{\rho}{2}||Ax - z - b||_2^2$$

According to Scaled ADMM algorithm we derived in (2.1), and assuming $A^T A$ is invertible, x^{k+1} for least absolute deviations will be:

$$x^{k+1} \leftarrow \arg\min_{x \in R^n}(\frac{\rho}{2}||Ax^k - z^k - b + u^k||_2^2) \tag{2.5}$$

we see this is a Least-Squares problem. So we have:

$$x^{k+1} = (A^T A)^{-1} A^T (b + z - u^k) \tag{2.6}$$

Also, z^{k+1} for least absolute deviations will be:

$$z^{k+1} \leftarrow \arg\min_{z \in R^n}(||z||_1 + \frac{\rho}{2}||Ax^{k+1} - z - b + u^k||_2^2) \tag{2.7}$$

To get the argmin, we have the gradient equation to solve:

$$\begin{aligned} 0 &= \nabla_z(||z||_1 + \frac{\rho}{2}(Ax^{k+1} - z - b + u^k)'(Ax^{k+1} - z - b + u^k)) \\ &= \nabla_z(||z||_1) + \frac{\rho}{2}\nabla_z(2x^{Tk+1}A^T z + z^T z + 2b^T z - 2u^{Tk} z) \\ &= \nabla_z(||z||_1) - \rho A x^{k+1} - \rho b + \rho u^k \end{aligned} \tag{2.8}$$

Thus we have

$$\begin{aligned} &\nabla_z(||z||_1) + \rho z = \rho A x^{k+1} + \rho z + \rho b + \rho u^k \\ &\frac{1}{\rho}\nabla_z(||z||_1) + z = A x^{k+1} - b + u^k \end{aligned} \tag{2.9}$$

there are three cases:

$$z^{k+1} = \begin{cases} 0 & \text{if } |Ax^{k+1} - b + u^k| \leq \frac{1}{\rho} \\ Ax^{k+1} - b + u^k - \frac{1}{\rho} & \text{if} \quad Ax^{k+1} - b + u^k > \frac{1}{\rho} \\ Ax^{k+1} - b + u^k + \frac{1}{\rho} & \text{if} \quad Ax^{k+1} - b + u^k < -\frac{1}{\rho} \end{cases}$$

After that, we update u^{k+1} by the linear equation

$$u^{k+1} = u^k + Ax^{k+1} - z^{k+1} - b$$

and from that, along with iteration (2.6) and (2.7) completes the ADMM for LAD.

3 Conclusion

From the two examples in previous section we can see ADMM explicitly targets problems that split into two distinct parts, f and g, that can then be handled separately. Problems of this form are so useful and common in machine learning, because by iteratively updating the parameters, computer can quickly minimize a loss function together with a regularization term or side constraints.

Bibliography

[1] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. *Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers*. Stanford University, 2011.